Real Time Rendering Iridescent Colors Appearing on Soap Bubbles

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Abstract. We present a novel technique for real-time rendering of iridescent colors appearing on soap bubbles. Previous techniques are based on multi-stage ray tracing techniques, or depend on user specified manual textures, which are unsuitable for real-time dynamic environment such as games. To render iridescent colors appearing on soap bubbles, we deploy optical phase reflectance based on light polarized perpendicular and parallel to the incident plane. On this framework, we implement intersection of the primary rays using rasterization, and efficiently approximate refraction and reflection for real-time performance. In addition, we simulate sloshing effects between the soap film using GPU based Perlin noise. This makes it possible to render the soap bubbles with visually convincing results in real-time dynamic environment.

1 Introduction

Color is the visual perceptual property corresponding in humans, and physical specifications of color are also associated with objects, materials, light sources etc. The color of an object is a complex result of its surface, transmission and emission properties, and is classified as original color and structural color. Original color is pigment color on the object. The structural colors, which are caused by optical path differences of reflected rays in microstructures. For example, iridescent colors of soap bubbles are especially famous. Since soap bubbles have beautiful structural color induced by light interference, they are often used for children's enjoyment, also used in soap bubble show. Therefore, the methodology to render these fascinating interference colors of soap bubbles has been an interesting research topic on graphics community.

Several methods have been developed to render the soap bubble with light interference[1, 2]. Since these methods employed a traditional ray tracing algorithm to render the soap bubbles, the computational cost is expensive. Multi-stage ray-tracing algorithm has been introduced, and uses multiple reflection and refraction for light interference[3]. This method creates realistic images even with large incident angles, it cannot be used in real-time application.

Kuck's technique gives appropriate approximation of ray-tracing and bubble dynamics[8]. But Kuck's model still cannot achieve real-time rendering performance. GPU based real-time rendering algorithm of soap bubble has been introduced[9].

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But this method cannot generate proper interference color. Durikovic introduces GPU based spectrum rendering[5], which shows real-time performance with limited colors.

GPU based real-time interference rendering algorithms with convincing results have been developed[4]. But this method ignores the refraction of soap bubbles that cause unnatural refraction effects. Also GPU based real-time structural color rendering technique has been developed[6]. Although this technique renders various structural colors including interference color of soap bubble, it highly depends on user specified manual textures. Therefore these techniques are not adequate for real-time dynamic environment such as game.

This paper proposes a new real-time rendering of iridescent colors appearing on soap bubbles without the user manual input. We combine optical phase reflectance and CIE-XYZ color transformations, and produce convincing results. Also our framework efficiently approximates refraction and reflection. This makes it possible to render the iridescent colors of soap bubbles with visually convincing result in real-time dynamic environment such as game. Our method's flow chart is shown in Figure 1.

Section 2 introduces how to accelerate the primary ray intersection using GPU rasterization. Section 3 presents the interference mapping according to optical phase reflectance and CIE-XYZ color transformation. Section 4 presents how to approximate refraction. Section 5 presents reflection mapping. Section 6 presents sloshing effect. Section 7 presents simulation results. Finally, the conclusion and future work are described in Section 8.

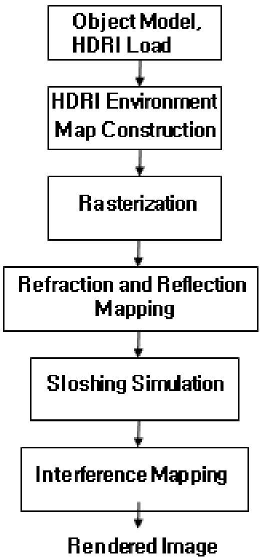

Fig. 1. Flow chart of the proposed method

2 Rasterization

The ray tracing performs many computations to find intersection points between the ray and the object surfaces in 3D space. To make reduce the computational overhead of finding the intersections with the primary ray, we use the rasterization method of the pipeline optimized for graphics hardware[13]. We adjust the projection setting as Equation (1) to match the results of the primary intersections by ray tracing and those of the rasterization. The projection angle of the rasterization is calculated efficiently from the relation between the camera and the image plane.

$$\mathrm{Pr}\,\mathit{ojection\ Angle} = 2 \times \tan^{-1}\left((\mathit{image\ plane\ width}/2)/\mathit{camera's\ z\ axis\ distance}\right) \tag{1}$$

3 Interference Mapping

Light have dual property, which are particle and wave property. One of the things that the wave model explain very well is interference, the source of soap bubble's Iridescent colors. Therefore we process light interference as spectrum based approach. We treat incident light represented as a sampled spectrum. Each sampled spectrum calculated by following two stages.

3.1 Optical Phase Reflectance Computation

Iridescent colors of soap bubble are very well visible and colorful in high radiance environment such as sunlight. Therefore we deploy optical equation for high radiance monochrome light and thin film in the optical society[17].

According this approach, if the film is illuminated by a light source with a spectral composition $I_{in}(\lambda)$, the reflected light intensity is given by Equation (2).

$$I_r(\lambda) = R_\lambda I_{in}(\lambda) \tag{2}$$

R_λ is reflectance, and given by Equation (3).

$$R_\lambda = R_s^2 \frac{1 - \cos\varphi}{1 + R_S^4 - 2R_S^2 \cos\varphi} + R_p^2 \frac{1 - \cos\varphi}{1 + R_p^4 - 2R_p^2 \cos\varphi} \tag{3}$$

R_s and R_p are the amplitude coefficients of reflection for light polarized perpendicular and parallel to the plane of incidence, and calculated by Equation (4).

$$R_s = \frac{\cos\theta - \sqrt{n^2 - \sin^2\theta}}{\cos\theta + \sqrt{n^2 - \sin^2\theta}}, \quad R_p = \frac{n^2\cos\theta - \sqrt{n^2 - \sin^2\theta}}{n^2\cos\theta + \sqrt{n^2 - \sin^2\theta}} \tag{4}$$

θ is the incident angle of the incident light. The phase difference between the reflected waves is given by Equation (5).

$$\varphi(\lambda, \theta) = \frac{4\pi d}{\lambda}\sqrt{n^2 - \sin^2\theta} \tag{5}$$

n is the refractive index of the soap film and d is the thickness of the soap film and λ is the wavelength of the incident light.

3.2 Spectrum Color Transformation

After computing intensity of reflectance, convert each wave's intensity to CIE-XYZ color to make appropriate color about light's wave property. First we compute Fourier basis function and incident wave spectrum[10]. And we compute CIE-XYZ color transformation using integration about plotted color matching function value[7]. Figure 2 shows color matching function and Equation (6) shows integration formula. Finally we convert XYZ color to RGB color and clamp range from zero to one.

$$\begin{aligned} X &= \int_0^\infty I(\lambda)\,\overline{x}(\lambda)\,d\lambda \\ Y &= \int_0^\infty I(\lambda)\,\overline{y}(\lambda)\,d\lambda \\ Z &= \int_0^\infty I(\lambda)\,\overline{z}(\lambda)\,d\lambda \end{aligned} \tag{6}$$

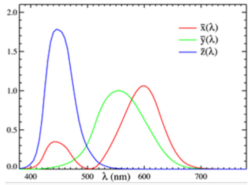

Fig. 2. Color matching function

4 Refraction Mapping

When the light participating other medium, the light refract light path according to Snell's law[11]. Figure 3 show this relation. Refraction vector can be derived from Snell's law, equation (7) shows refraction vector according to refractive index.

$$\begin{aligned} &\underline{r} = \left(-\eta(\underline{N}\bullet\ \underline{I}) - \sqrt{\eta^2(\underline{N}\bullet\ \underline{I})^2 - \eta^2 + 1}\right)\underline{N} + \eta\underline{I} \\ &\text{if } \eta < 1 \\ &\underline{r} = \left(-\eta(\underline{N}\bullet\ \underline{I}) + \sqrt{\eta^2(\underline{N}\bullet\ \underline{I})^2 - \eta^2 + 1}\right)\underline{N} + \eta\underline{I} \\ &\text{if } \eta > 1 \end{aligned} \tag{7}$$

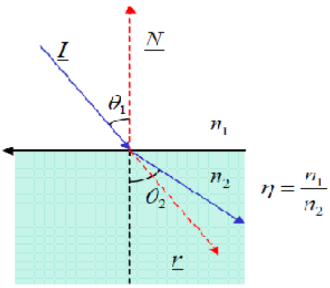

Fig. 3. Refraction of Light

If exit point from inside to outside is computed by intersection test, computation cost is very heavy. Since single soap bubble has four time refraction, computation cost is more important problem. Therefore we efficiently approximate this process by deploying Wyman's approximation technique[15].

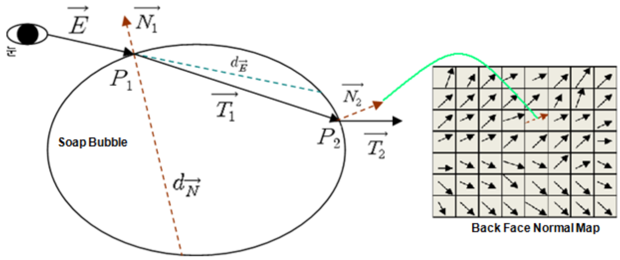

Fig. 4. Refraction approximation using back face normal map

Figure 4 shows briefly refraction approximation. In the eye position, we find difference of depth value($d_{\vec{E}}$) of front face and back face about soap bubble. And produce normal map of back face of soap bubble. Then get the difference of depth value ($d_{\vec{N}}$) about front face and back face in the position of $P_1 + \vec{N_1}$. And get the $\vec{T_1}$ from $\vec{E}$ and $\vec{N_1}$ using Equation (7).

$$P_2 = P_1 + \tilde{d}\,\overrightarrow{T_1} \qquad d = \left(d_{\vec{E}} + d_{\vec{N}}\right)/2 \tag{8}$$

We get the light exit position(P_2) from inside to outside of soap bubble using equation (8). P_2 is multiplied by projection matrix defined at the eye position, project to back face normal map. Then we get normal vector $\vec{N_2}$ at the exit position and get the refraction vector $\vec{T_2}$. Finally $\vec{T_2}$ is used for refraction from environment map.

5 Reflection and HDRI Mapping

Generally fresnel's term is widely used to represent reflection effect on the surface of transparent object such as soap bubble. But fresnel's term have relatively computational cost. Therefore we use approximation method of fresnel's term to increase more rendering performance[12]. Equation (9) shows this approximation term and θis the angle between incident light and normal vector about refractive surface.

$$R(\theta) = R_0 + (1 - R_0)(1 - \cos\theta)^5 \tag{9}$$

Traditional imaging systems have limitation to represent the radiance distribution of real world. HDRI(High Dynamic Range Image) was innovative technique to overcome this limitation. To represent the radiance of real world, We use real-time HDRI texture mapping technique to construct HDR environment cube map[14]. Therefore radiance of natural lights are reflected and refracted from soap bubble by using environment mapping technique. Figure 5 show HDRI environment mapping.

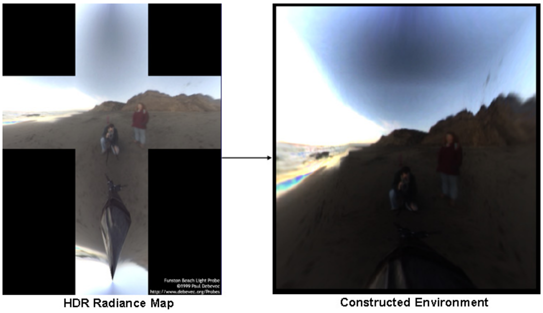

Fig. 5. HDRI Environment Mapping

6 Sloshing Simulation

Soap bubble is composed of soap molecules and water molecules. In a soap film, two parallel sheets of soap molecules sandwich a layer of water molecules. Because water molecules move well relative soap molecules by external force such as wind and gravity, soap bubble have sloshing effect due to the water molecules movement. We approximate sloshing effect by varying the thickness of moving soap bubble using GPU based Perlin Noise technique[16]. Sloshing effect calculated on fragment program about each thickness of fragment.

7 Simulation Results

We used a computer equipped with Intel i7 CPU 2.80Ghz, Nvidia GTX260 graphics hardware. Resolution is 1024x1024. Development environment are Visual C++ based on Window7, OpenGL, and GLSL. Each soap bubble is composed of sphere model that have 10,242 vertices and 20480 triangles.

Figure 6 shows two soap bubbles rendering results. Total 40,960 triangles rendered average 76.1 frame per second. And Figure 7 show 56 soap bubbles rendering results. Total 1,146,880 triangles rendered average 24.5 frame per second in Figure 7.

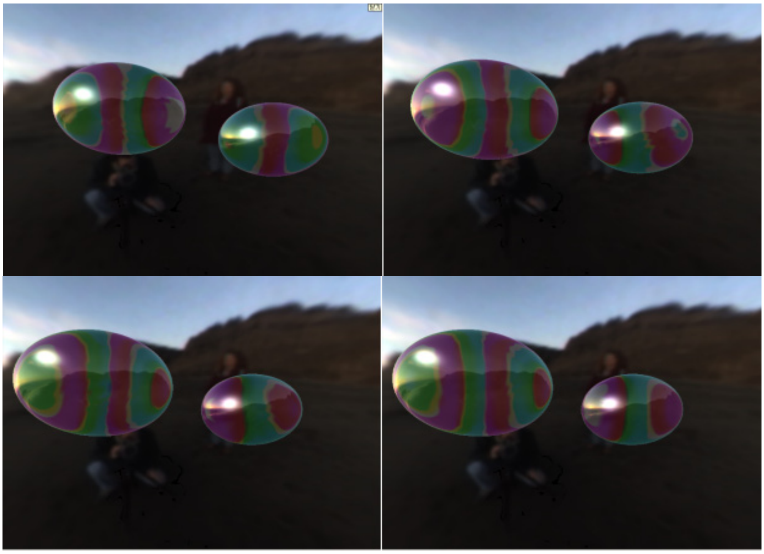

Fig. 6. Two soap bubble rendering result

Fig. 7. Total 56 soap bubbles rendering

8 Conclusion

This paper presents a real-time rendering of soap bubbles considering light interference. We render iridescent colors appearing on soap bubbles using optical phase reflectance about polarized to the incident plane and CIE-XYZ color transformation. In order to improve the rendering performance, we include several methods, which are GPU based rasterization method and efficient approximation of refraction and reflection. And we simulate sloshing effect between the soap film using GPU based Perlin Noise. The proposed system provides convincing rendering result of the soap bubbles considering light interference in real-time dynamic environment. For future work, we would like to simulate draining phenomena and implement more accurate interference effect.

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